

Quang Truong

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Professional Summary

Ph.D. student with 4+ years of experience in developing scalable Deep Learning (DL) solutions. Proficient in major DL frameworks such as Pytorch and Tensorflow. Strong software development skills with 7 years of experience in Python and C/C++ with prior records of open-source contributions. A proven track record of publishing and reviewing research papers in renowned journals and conferences such as AAAI, JMLR, TKDD, ICLR, and NeurIPS.

Education

Michigan State University

Ph.D. in Computer Science

East Lansing, MI

Aug. 2024 - Present

Dartmouth College

M.S. in Engineering Sciences

Hanover, NH

Sep. 2022 - Jun. 2024

Selected Publications

‡ indicates equal contribution.

Preprints

- [1] A Pre-training Framework for Relational Data with Information-theoretic Principles
Quang Truong, Zhikai Chen, Mingxuan Ju, Tong Zhao, Neil Shah, and Jiliang Tang. **2025a**
Under review at NeurIPS'25. arXiv: 2507.09837 [cs.LG].
- [2] Position: Structural Understanding and Intrinsic Reasoning Enable Large Language Models to Perform Graph Reasoning
Xinnan Dai, Caihua Shan, **Quang Truong**, Haoyu Han, Yuxiang Wang, Yao Ma, Dongsheng Li, and Jiliang Tang. **2025b**
Preprint.

Peer-reviewed Publications

- [3] TopoX: A Suite of Python Packages for Machine Learning on Topological Domains
Mustafa Hajji‡, Mathilde Papillon‡, Florian Frantzen‡, ... **Quang Truong**, and ... Nina Miolane. **2024a**
The Journal of Machine Learning Research (Volume 25).
- [4] Weisfeiler and Lehman Go Paths: Learning Topological Features via Path Complexes
Quang Truong and Peter Chin. **2024b**
AAAI'24: The 38th Annual AAAI Conference on Artificial Intelligence.
- [5] Not All Data Matters: An Efficient Approach to Multi-Domain Learning in Vehicle Re-Identification
Quang Truong and Bo Mei. **2021**
ITSC'21: The 24th IEEE International Conference on Intelligent Transportation Systems.

Open-source Projects

TopoX

- Provides reliable and user-friendly building blocks for computing and machine learning on topological domains that extend graphs.
- Starred 500+ times across 3 different libraries on GitHub.
- Published at the Journal of Machine Learning Research (JMLR).

Skills

Programming	Python (Expert), C/C++ (Advanced), Java (Proficient), R (Proficient), MATLAB (Proficient)
Machine Learning	Graph Machine Learning, Relational Deep Learning, Knowledge Graphs, Recommendation System, Computer Vision, Natural Language Processing, Distributed Data Parallel
ML Frameworks	Pytorch, Pytorch Geometric, Pytorch Frame, NetworkX, TopoX, Pytorch Lightning, LangChain, Hugging Face, Scikit-learn, Tensorflow, Keras, Matplotlib, Numpy, Pandas
DevOps	WandB, Docker, Singularity, Pytest, Amazon Web Services (AWS), Google Cloud Platform (GCP)
Software Engineering	Spring, Agile, JIRA, SCRUM, Git, Linux/Unix, MongoDB, DuckDB, MySQL

Research Experience

DSE Lab, Michigan State University

Doctoral Research Assistant (Advisor: **Jiliang Tang**)

East Lansing, MI

Aug. 2024 - Present

- Investigate the expressive capacity of decoder-only Transformer architectures for solving graph connectivity problems.
- Developed specialized pre-training procedures for graph machine learning on relational databases, thereby enhancing predictive modeling for node-level classification, link prediction, and temporal forecasting tasks (**Snap Research Collaboration**).

LISP Lab, Dartmouth College

Graduate Research Assistant (Advisor: **Peter Chin**)

Hanover, NH

Sep. 2022 - Jun. 2024

- Introduced a novel approach to Topological Deep Learning (TDL) leveraging simple paths, significantly enhancing graph expressivity and surpassing the theoretical limitations of the 3-WL test.
- Developed Path Isomorphism Network, a state-of-the-art TDL model, which delivered outstanding performance on benchmark datasets: PROTEINS (78.8%), NCI1 (85.1%), NCI109 (84.0%), and IMDB-B (76.6%).
- Developed an anomaly detection pipeline for human behavior analysis using geospatially-tagged data, as part of the **IARPA-HAYSTAC** funded project.

Vision Lab, University of Illinois at Chicago

Research Intern (Advisor: **Wei Tang**)

Remote

May 2021 - Aug. 2021

- Designed a context-aware 3D object detection architecture that captures both object-wise and object-context relationships, improving spatial understanding.
- Boosted model performance through cross-modality alignment, improving the feature representation of objects.

ML Lab, Texas Christian University

Undergraduate Research Assistant (Advisor: **Bo Mei**)

Fort Worth, TX

May 2019 - May 2021

- Developed an advanced Vehicle Re-identification pipeline utilizing Generative Adversarial Networks (GANs) for adaptive domain learning, resulting in a 10% improvement in mean Average Precision (mAP) over the baseline.
- Engineered a novel filtering algorithm that reduces the training data requirement by 25% while maintaining model performance, significantly improving computational efficiency.

Presentations

Invited Presentations

- Topologically-driven Neural Networks, *HPCC Special Seminar: Decoding Computational Challenges, the Advanced Institute of Interdisciplinary Science and Technology - HCMUT, VNU-HCM*. **2024**

Conference Presentations

- Weisfeiler and Lehman Go Paths: Learning Topological Features via Path Complexes, *AAAI'24: The 38th Annual AAAI Conference on Artificial Intelligence*. **2024**
- Not All Data Matters: An Efficient Approach to Multi-Domain Learning in Vehicle Re-identification, *ITSC'21: The 24th IEEE International Conference on Intelligent Transportation Systems*. **2021**

Professional Services

Reviewer

- Conference Reviewer, *Conference on Neural Information Processing Systems (NeurIPS)*. **Summer 2025**
- Journal Reviewer, *IEEE Transactions on Big Data (TBD)*. **Summer 2025**
- Conference Reviewer, *International Conference on Learning Representations (ICLR)*. **Fall 2024**
- Journal Reviewer, *ACM Transactions on Knowledge Discovery from Data (TKDD)*. **Fall 2024**

Teaching Experience

- Teaching Assistant, *ENGS 96: Mathematics for Machine Learning, Dartmouth College*. **Winter 2024**

Honors

- VEF Fellow, *VEF 2.0 Program, VEF Fellows and Scholars Association*. **2021**
- Vingroup Scholarship Recipient, *Master's & Ph.D. Scholarship Program, VinUniversity*. **2021**
- Bronze Medal, *ACM-ICPC South Central USA Regional Contest [URL]*. **2019**